Cisco PE Node Failure SOP - Routing Protocol Flap (MALL-PE-08)

Node: MALL-PE-08 | **Vendor:** Cisco | **Severity:** Minor | **Alarm Code:** ALARM_3213

Description

The Malleswaram PE Node (MALL-PE-08) failure due to a routing protocol flap occurs due to an underlying instability in the BGP routing protocol, which manifests as a repetitive route churn, or alternatively, an OSPF adjacency loss and ISIS LSP flooding. The PE node, being a critical component of the PE-CE routing hierarchy, plays a pivotal role in forwarding and routing packets between the customer edge and the provider edge networks. The Cisco IOS software, running on the PE node, utilizes the BGP protocol to establish and maintain routing sessions with various neighboring Autonomous Systems (AS). However, due to various reasons such as configuration inconsistencies, link failures, or software bugs, the BGP protocol may become unstable, resulting in a routing protocol flap. This instability can further propagate to other protocols such as OSPF and ISIS, leading to adjacency losses and LSP flooding, ultimately impacting the overall network performance and reliability. Therefore, it is essential to identify and rectify the root cause of the BGP protocol instability to prevent further network disruptions.

Pre-requisite Checks

- check: Verify the power supply and cooling systems at the Malleswaram Node
- check: Check the redundant link status between the PE node and the CE node
- check: Confirm the software version and configuration of the PE node

Potential Root Causes

Cause	Probability
Misconfigured BGP neighbor	High
BGP route reflector misconfiguration	Medium
Link failure between PE and CE nodes	Low

Diagnostic Commands & Expected Outputs

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Command: show ip bgp summary, Expected Output: BGP table version is 3, local router ID is 192.168.1.1, Description: This command displays a brief summary of the BGP routing table, including the number of peer connections, the number of received updates, and the number of active routes. By analyzing the output, we can determine if there are any issues with the BGP routing table or if the BGP protocol is stable.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Interface IP Address, Description: This command displays information about the OSPF neighbors, including the neighbor ID,
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priority, state, dead time, and interface. By analyzing the output, we can determine if there are any OSPF adjacency losses or if the OSPF protocol is stable.

Command: show isis adjacency, Expected Output: IS-IS adjacency database, Description: This command displays information about the ISIS adjacency database, including the adjacency state and the interface. By analyzing the output, we can determine if there are any ISIS adjacency losses or if the ISIS protocol is stable.
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Resolution Steps

Step 1: Step: Verify and correct the BGP neighbor configuration, Command: show run | include bgp neighbor, Description: Verify the BGP neighbor configuration on the PE node and correct any misconfigurations.

Step 2: Step: Verify and correct the BGP route reflector configuration, Command: show run | include bgp route-reflector, Description: Verify the BGP route reflector configuration on the PE node and correct any misconfigurations.

Step 3: Step: Verify and repair the link between the PE and CE nodes, Command: show ip int brief, Description: Verify the link status between the PE and CE nodes and repair any link failures.

Post-Check Verification

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Command: show ip bgp summary, Expected Output: BGP table version is 3, local router ID is 192.168.1.1, Description: Verify that the BGP routing table is stable and there are no issues with the BGP protocol.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Interface IP Address, Description: Verify that the OSPF protocol is stable and there are no OSPF adjacency losses.

Command: show isis adjacency, Expected Output: IS-IS adjacency database, Description: Verify that the ISIS protocol is stable and there are no ISIS adjacency losses.
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Roll-back Procedures

- Step: Restore the previous BGP neighbor configuration, Command: copy run start, Description: Restore the previous BGP neighbor configuration by copying the running configuration back to the startup configuration.

- Step: Restore the previous BGP route reflector configuration, Command: copy run start, Description: Restore the previous BGP route reflector configuration by copying the running configuration back to the startup configuration.

- Step: Restore the previous link configuration, Command: copy run start, Description: Restore the previous link configuration by copying the running configuration back to the startup configuration.